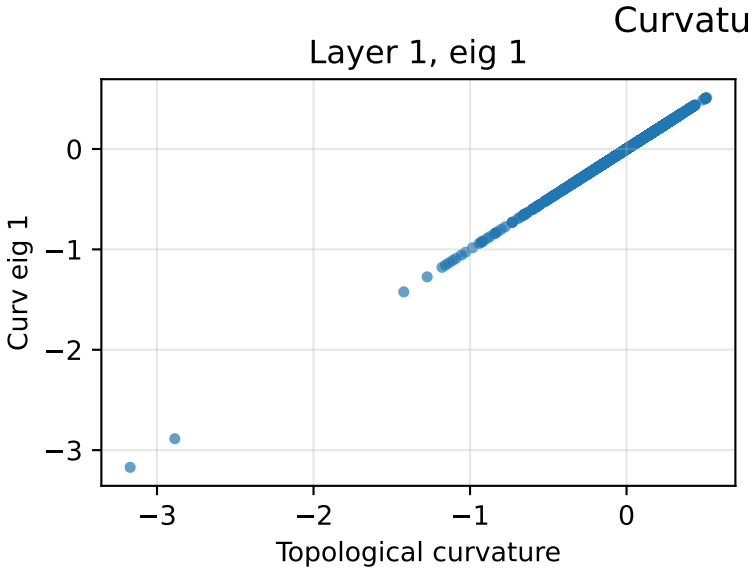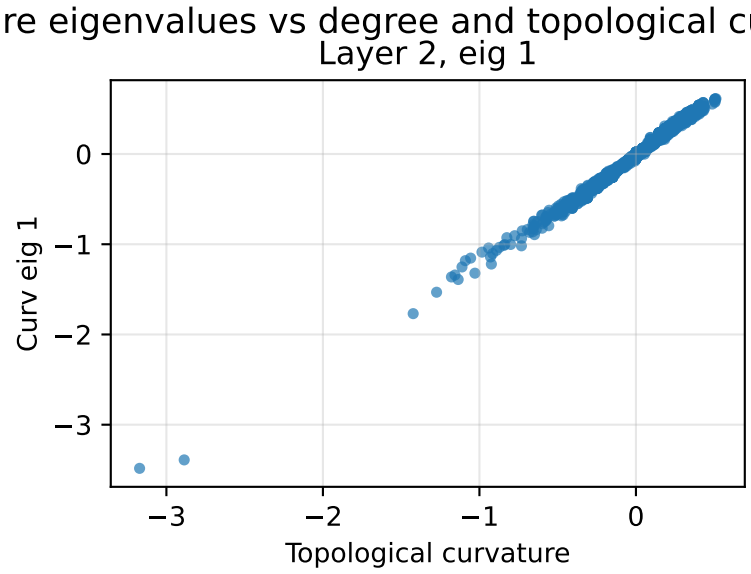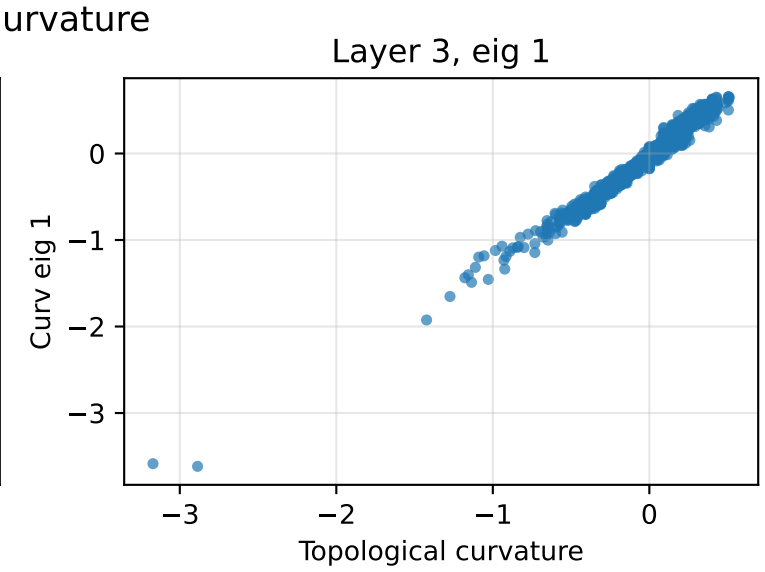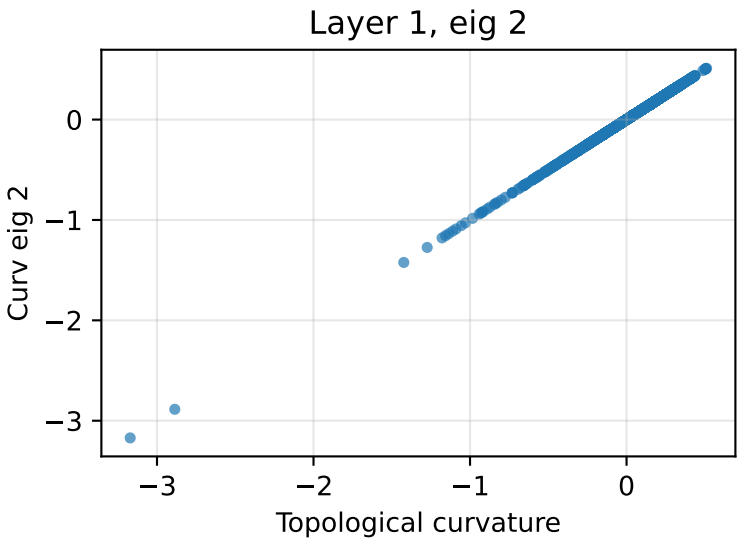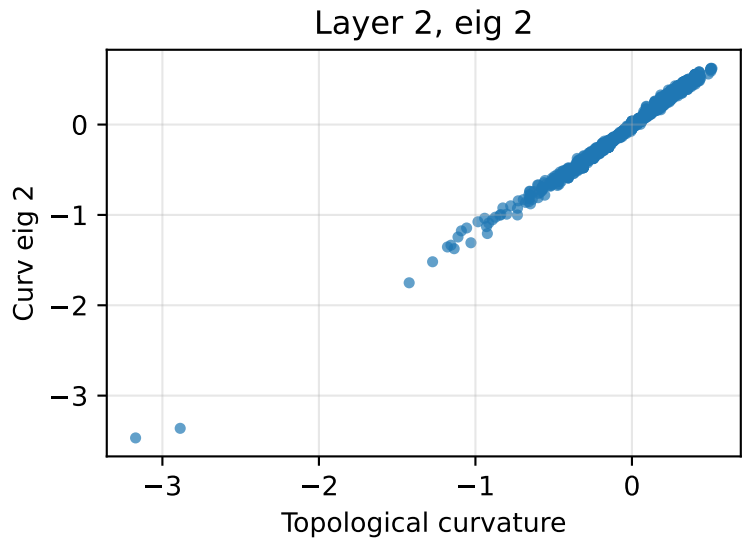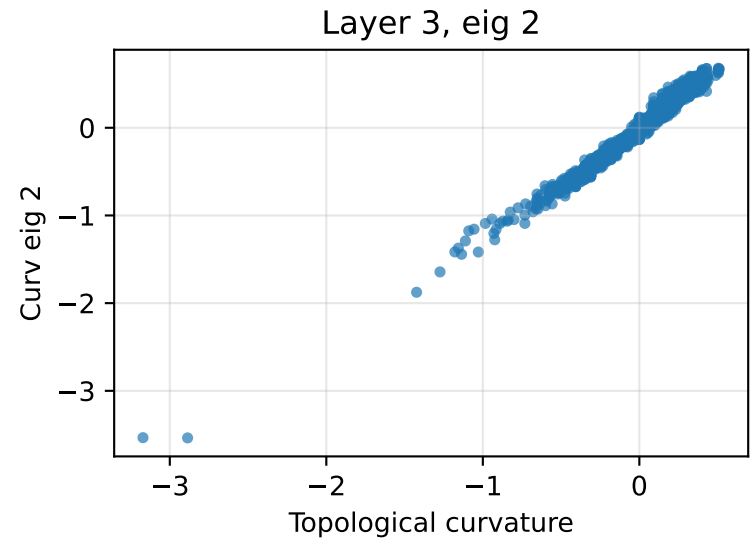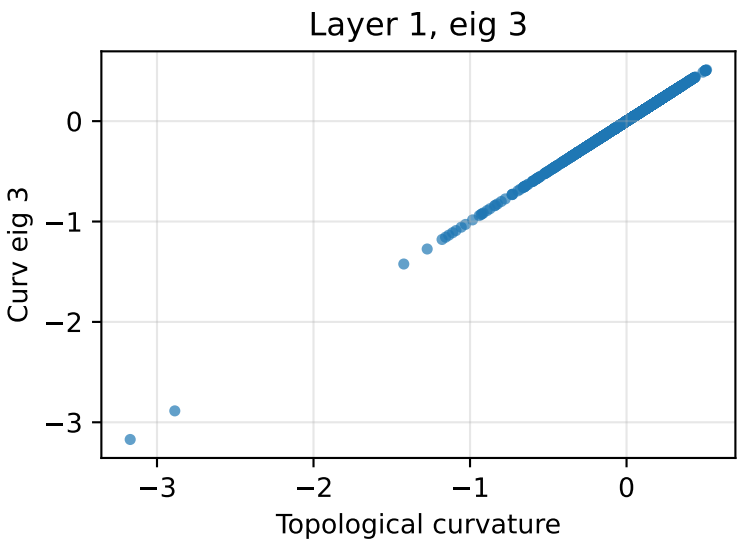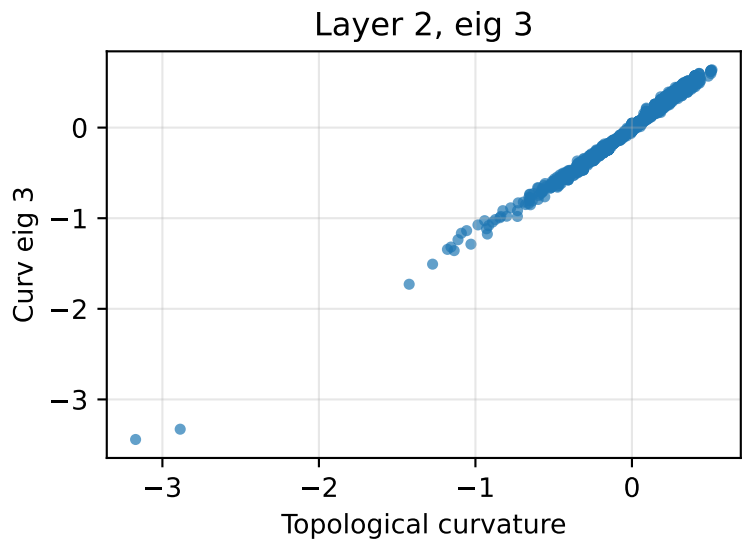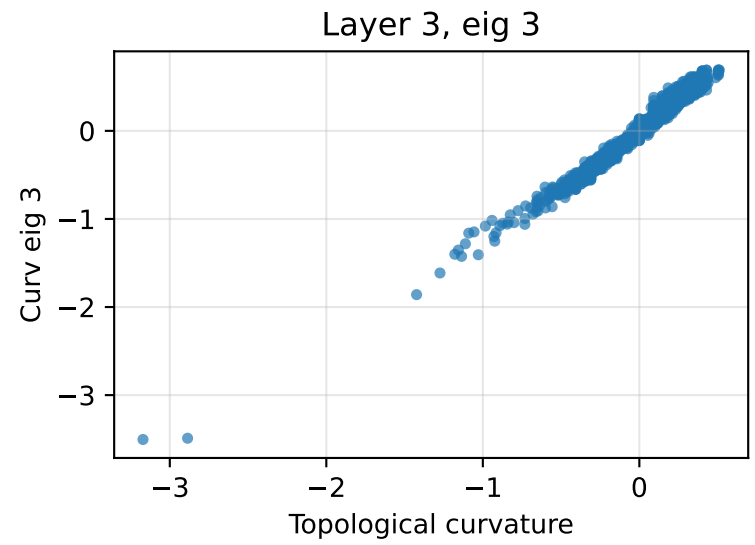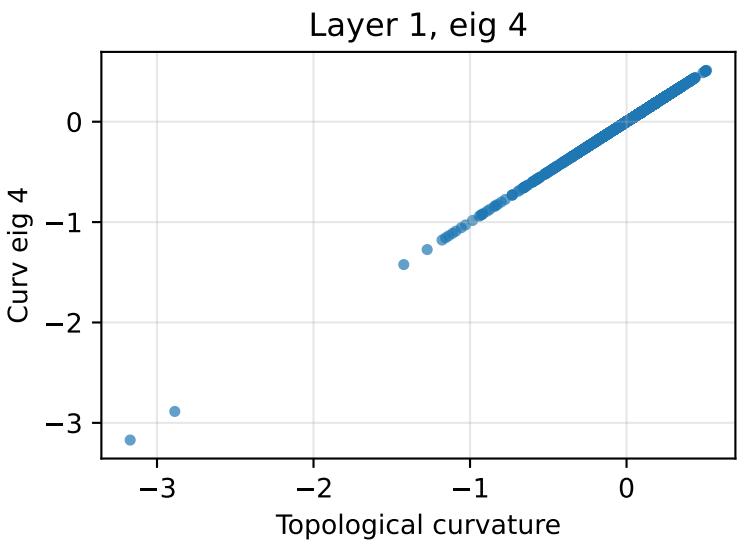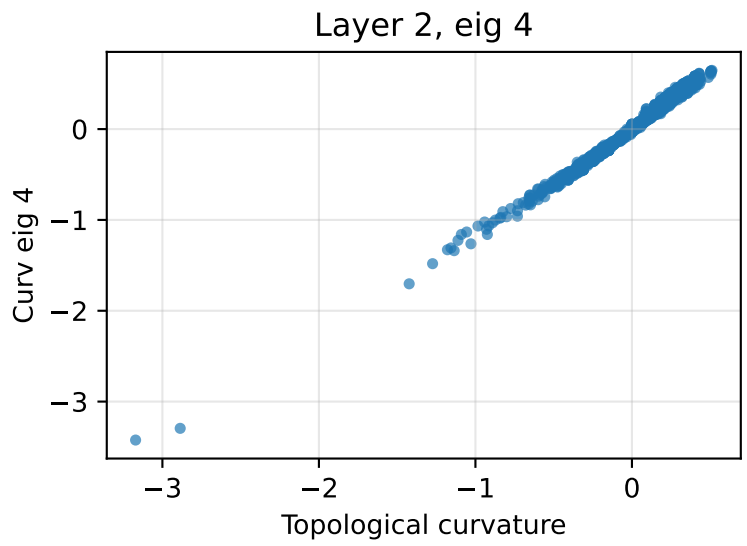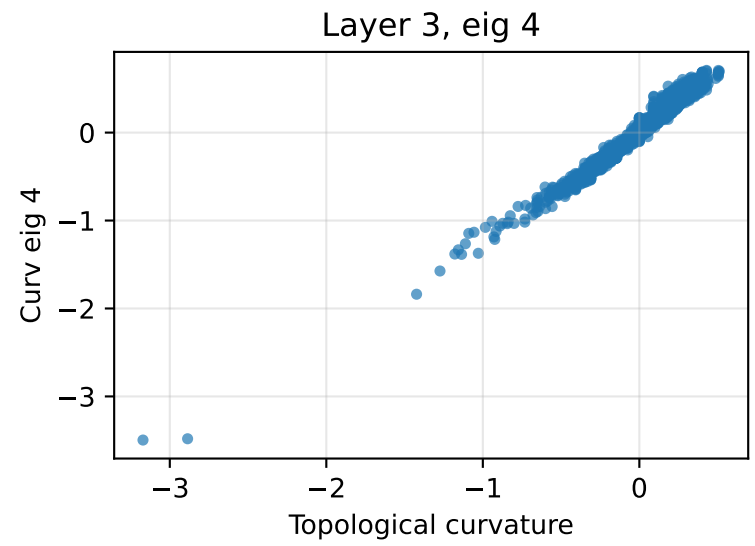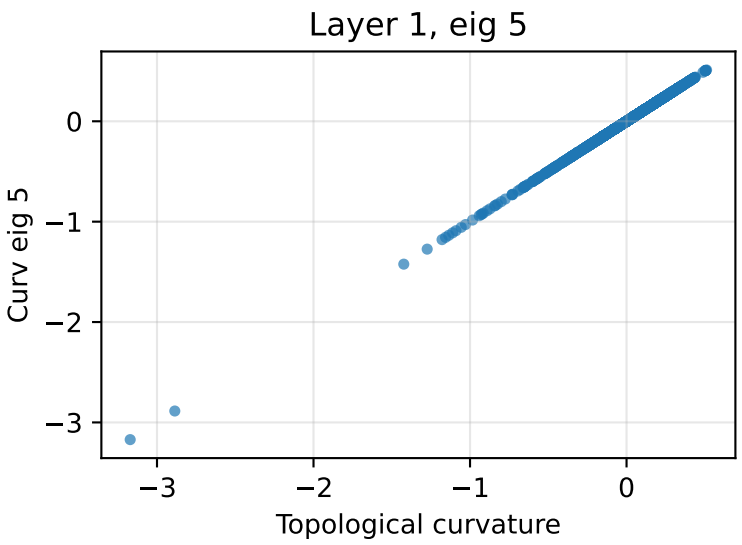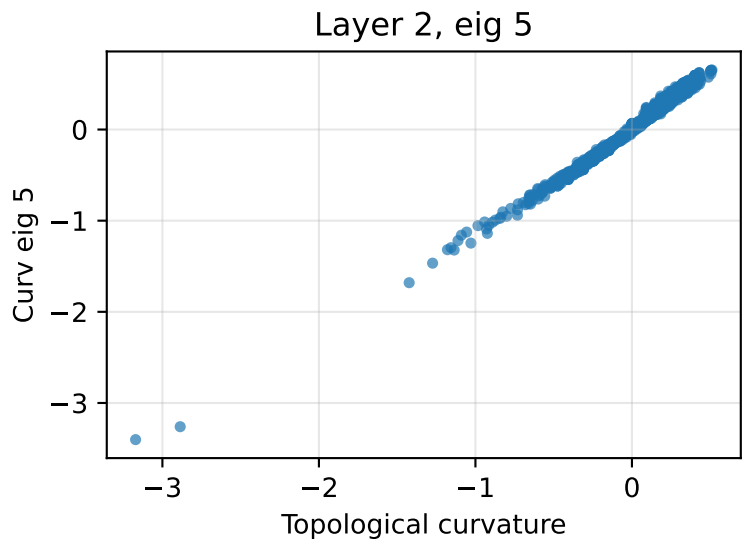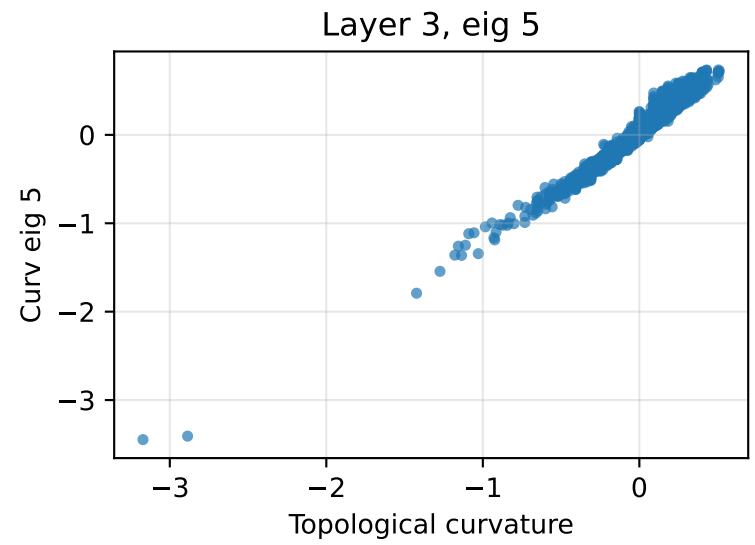Curvature eigenvalues vs degree and topological curvature